

Shruti Kumari

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SUMMARY

Data Scientist / ML Engineer with production experience across predictive maintenance, anomaly detection, and multimodal RAG, with measurable impact in downtime reduction, detection accuracy, and inference latency. Comfortable owning the full stack from feature engineering, model training, explainability (SHAP), and deployment (Docker, FastAPI, Kubernetes).

EDUCATION

University at Buffalo, The State University of New York, USA

Master of Science, Computer Science

WORK EXPERIENCE

University at Buffalo

Machine Learning Research Assistant

July 2026 - Present

- Built a Multimodal RAG Pipeline for Automated Radiology Report Generation on MIMIC-CXR dataset.
- Used BioMedCLIP (ViT-B/16) to embed images, enable retrieval via FAISS and a multi-phase generation module.
- Delivered findings and impressions benchmarked against ground truth using BLEU, ROUGE-L, BERTScore, and CheXbert F1.

Hewlett Packard Enterprise

Software Engineer (Data Scientist)

Aug 2022 - Jul 2024

- Built a disk failure prediction model using Random Forest & Gradient Boosting, achieved 88% precision on storage data.
- Engineered 40+ time-series features from telemetry data, reduced unplanned downtime by 30% across storage clusters.
- Improved Mean Time to Detect (MTTD) by 25% via predictive modeling integrated with REST APIs into production workflows.
- Built anomaly detection platform using LSTM and Transformer models on Kafka streams, cutting false positives by 50%.
- Designed end-to-end pipeline with cross-validation, model retraining, and PostgreSQL result storage for hardware forecasting.
- Accelerated operational MTTR by 30% by shipping Grafana dashboards on Kubernetes for real-time health monitoring.
- Collaborated with cross-functional teams to align ML model outputs with operational incident workflows.

Research & Development Intern (Data Science Intern)

Jan 2022 - Jul 2022

- Delivered a Scikit-learn bug prediction model integrated with Bugzilla, achieved 85% accuracy across 100+ projects.
- Enhanced triage accuracy by 25% by building NLP feature extraction pipeline on unstructured bug reports.
- Improved bug matching time by 70% by containerizing prediction service with Docker and enabled reproducible retraining.

PROJECTS

Credit Card Retention Prediction System with Explainable AI Dashboard | [Link](#)

- Developed an end-to-end churn prediction pipeline on 10k+ customers using XGBoost, Random Forest, Logistic Regression.
- Engineered domain-informed features from 21 raw inputs, confirmed with SHAP TreeExplainer, validated feature design.
- Handled 16% class imbalance with and without SMOTE, ran stratified 5-fold cross-validation on training data.
- Deployed a 5-page Streamlit dashboard with live SHAP waterfall explanations and business ROI calculator.

NLP Sentiment & Topic Intelligence Dashboard for Amazon Product Reviews | [Link](#)

- Designed and compared VADER and DistilBERT sentiment pipelines on 24k+ Amazon reviews, with agreement tracking.
- Built and tuned LDA topic model across 21k+ documents to find coherent topics with validated labels.
- Delivered findings through a deployed 4-page Streamlit cloud dashboard with Plotly charts.

Clinical Readmission Risk Prediction with Explainability Report | [Link](#)

- Processed the UCI diabetes readmission dataset (100K+ rows) with SQL profiling queries.
- Applied SMOTE for class imbalance, trained a gradient-boosted classifier, achieved strong predictive performance.
- Validated via chi-square and t-test hypothesis testing on key risk factors.
- Implemented SHAP for model explainability, generated a clinical decision report, confusion matrix and feature importance plots.

SKILLS

- **Languages & Data Tools:** Python, R, SQL, PostgreSQL, Pandas, NumPy
- **Machine Learning & Statistics:** Scikit-learn, XGBoost, Random Forest, Gradient Boosting, LSTM, Transformers, Statistical Modeling, Hypothesis Testing, Time-Series Analysis, Feature Engineering, SMOTE, Cross-Validation
- **AI & NLP:** PyTorch, TensorFlow, DistilBERT, BERTopic, Sentiment Analysis, Topic Modeling, SHAP
- **Data Visualization & Analytics:** Tableau, Plotly, Seaborn, EDA, Data Visualization, ROC-AUC, BLEU, ROUGE
- **Cloud, MLOps & Tools:** AWS, Docker, MLflow, FastAPI, Git